

Tide retrospective: local uniform agreement

a Lipschitz peak turns L^1 decay into L^∞ decay

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2026-09-19

Abstract

For smooth nonnegative losses whose Laplace families agree beyond all orders on smooth compactly supported tests, the Boltzmann weights themselves agree uniformly on compacts beyond all orders: for every compact K and every N there is C with $|e^{-tL_2(x)} - e^{-tL_1(x)}| \leq C t^{-N}$ for all $x \in K$ and all large t , and in particular the weights agree beyond all orders at every point. The input is the local total-variation decay of the previous tide; the mechanism is a peak argument. The weights take values in $[0, 1]$ and have gradient of norm at most $t \|\nabla L\|$, so their difference is Λ -Lipschitz on unit balls with $\Lambda = tG + 1$; a peak of height H therefore carries mass at least $H/2$ on a ball of radius $H/(2\Lambda)$, whence $H^{d+1} \leq (2^{d+1}/c_d) \Lambda^d \int_{K'} |e^{-tL_2} - e^{-tL_1}|$ on the unit thickening K' of K . The powered inequality is stated for any bounded Lipschitz function and is the reusable part. Three hundred and forty lines, three LSP rounds.

1 Setting

The total-variation tide (2026-09-19-17-00) proved $\int_K |e^{-tL_2} - e^{-tL_1}| = o(t^{-\infty})$ on every compact K , and its consult remarked that the local L^∞ version is also true under smoothness, sketching a gradient bound and an interpolation. This tide formalises that remark. The deliberation (tide log 2026-09-19-18-30-tide-germbij-local-uniform.md in tide-log/, consult in gpt_germbij_lu_v1.md) confirmed the exponent $d + 1$ and the constant, asked for the degenerate cases ($H = 0$, $G = 0$, empty K) to be handled explicitly, recommended the uniform-bound API over a real supremum, and recommended stating the peak lemma generically. Both parties voted for the package.

2 The thing formalised

Write $h_t = e^{-tL_2} - e^{-tL_1}$, d for the dimension, c_d for the volume of the unit ball, and $K' = \overline{K}_1$ for the closed 1-thickening of K .

(P) Let f be integrable on K' , with $|f(x_0)| \leq 1$, $B(x_0, 1) \subseteq K'$, and $|f(y) - f(x_0)| \leq \Lambda \|y - x_0\|$ on $B(x_0, 1)$ for some $\Lambda \geq 1$. Then $|f(x_0)|^{d+1} \leq (2^{d+1}/c_d) \Lambda^d \int_{K'} |f|$.

(W) For $L_1, L_2 \geq 0$ smooth and K compact there is C with $|h_t(x)|^{d+1} \leq C t^d \int_{K'} |h_t|$ for all $t \geq 1$, $x \in K$; equivalently $|h_t(x)| \leq (C t^d \int_{K'} |h_t|)^{1/(d+1)}$.

(U) If moreover the families agree beyond all orders on smooth tests, then for every N there is C with $|h_t(x)| \leq C t^{-N}$ for all $x \in K$, eventually in t ; and $h_t(x) = o(t^{-\infty})$ for every x .

The Lean signature of (U):

```
theorem eventually_uniform_abs_exp_sub_le {L1 L2 : (iota -> R) -> R}
  (h1 : ContDiff R infty L1) (h2 : ContDiff R infty L2)
  (hL1 : forall w, 0 <= L1 w) (hL2 : forall w, 0 <= L2 w)
  (hexact : forall phi, ContDiff R infty phi -> HasCompactSupport phi ->
    SuperPoly fun t => I2 phi t - I1 phi t)
  {K : Set (iota -> R)} (hK : IsCompact K) :
  forall N : Nat, exists C : R, eventually t in atTop, forall x in K,
    |exp (-(t * L2 x)) - exp (-(t * L1 x))| <= C * t ^ (-(N : R))
```

(integrals abbreviated). The other declarations:

- (P): the peak lemma `pow_abs_le_of_lipschitz_of_setIntegral`;
- gradients: `hasFDerivAt_expWeight` and `norm_expWeight_deriv_le`;
- the difference gradient: `norm_fderiv_weightDiff_le`;
- the amplitude bound `abs_exp_sub_le_one` and the gradient bound on a compact, `exists_gradient_bound_on`;
- (W): `pow_abs_exp_sub_le` and `abs_exp_sub_le_rpow`;
- the pointwise corollary `superPoly_exp_sub_at`.

Nothing about the losses beyond differentiability with a bounded gradient near K is used in (P) and (W); the smoothness hypothesis is inherited from the upstream agreement hypothesis, which tested $\eta^2(L_2 - L_1)$.

3 Proof strategy

Gradients. $D(e^{-tL})(x) = e^{-tL(x)}(-tDL(x))$ by the chain rule (`HasFDerivAt.exp` after `const_mul` and `neg`), so $\|D(e^{-tL})(x)\| \leq t\|DL(x)\|$ for $t \geq 0$ since the weight is at most 1. The difference has $\|Dh_t\| \leq t(\|DL_1\| + \|DL_2\|) \leq tG$ on K' , with G from continuity of `fderiv` on the compact K' (`ContDiff.continuous_fderiv`, `IsCompact.exists_bound_of_continuousOn`).

Lipschitz on unit balls. The mean value inequality on the convex ball,

$$\text{Convex.norm_image_sub_le_of_norm_fderiv_le}$$

with `DifferentiableAt` hypotheses, gives $|h_t(y) - h_t(x_0)| \leq tG\|y - x_0\| \leq \Lambda\|y - x_0\|$. Taking $\Lambda = tG + 1$ rather than tG costs nothing and makes $\Lambda \geq 1$, which is what keeps the peak radius $H/(2\Lambda) \leq 1/2$ inside the unit ball, with no case distinction on G .

Peak. If $H = |f(x_0)| = 0$ there is nothing to prove. Otherwise on $B(x_0, r)$ with $r = H/(2\Lambda)$ one has $|f| \geq H - \Lambda r = H/2$, so `setIntegral_ge_of_const_le` and monotonicity of

the set integral give $\int_{K'} |f| \geq (H/2) \text{vol}(B(x_0, r))$, and `Measure.addHaar_ball_of_pos` with `Module.finrank_fintype_fun_eq_card` evaluates the volume as $r^d c_d$. The algebra $H^{d+1} = (r^d c_d \cdot H/2) \cdot (2^{d+1}/c_d) \Lambda^d$ is one `field_simp; ring`.

Assembly. For $t \geq 1$, $\Lambda \leq (G + 1)t$, so $|h_t(x)|^{d+1} \leq Ct^d \int_{K'} |h_t|$ with $C = (2^{d+1}/c_d)(G + 1)^d$. Feeding the total-variation decay at exponent $(d + 1)N + d$ gives $|h_t(x)|^{d+1} \leq C(t^{-N})^{d+1} = (C^{1/(d+1)}t^{-N})^{d+1}$ (`Real.rpow_inv_natCast_pow`) and `le_of_pow_le_pow_left0` takes the root. The pointwise corollary applies this to $K = \{x\}$ and the seabed’s `superPoly_of_forall_eventually_le`.

4 Roadblocks and resolutions

Three LSP rounds. The pointwise lemmas about e^{-tL} cannot omit the `Fintype` instance once a `fderiv` on $\iota \rightarrow \mathbb{R}$ appears (the `Pi` normed-group instance needs it), and `HasFDerivAt.sub` returns its function in `Pi` form, so the rewrite by its `fderiv` must go through a `have` with the `lambda` type stated. A `set H := {f x0 | }` placed before the case split on $H = 0$ left the hypothesis in the unfolded spelling; the split now comes first. An `abs_sub_le_iff` rewrite produced an `isDefEq` timeout for no visible reason and was replaced by `abs_le.mpr` with two `linarith` calls. `ring` does not distribute a symbolic power over a product, so `mul_pow` precedes it. The implicit compact set of the total-variation theorem must be named when the argument is a thickening.

The consult, before writing, fixed the shape and caught the scope of a claim in my own notes:

GPT-6 Astra: Derivatives: the proposed “so no” is not correct. Under C^∞ , all fixed-order spatial derivatives are locally uniformly superpolynomially small. [...] For a unit vector v and $0 < s \leq 1$, Taylor’s estimate along a segment gives $|Dh_t(x)v| \leq 2A_t/s + \frac{1}{2}B_K t^2 s$. For target exponent N , choose $s = t^{-(N+3)}$.

That is recorded as the next follow-up. On the API:

GPT-6 Astra: The uniform-bound formulation is a good primary Lean API. [...] The subtype, compact-space instance, and norm evaluation infrastructure are probably more work than they save in this tide.

5 What was Mathlib, what was new

Mathlib, all lookup:

- calculus: `HasFDerivAt.exp`, `HasFDerivAt.const_mul`, and `HasFDerivAt.sub`;
- regularity: `ContDiff.continuous_fderiv` and `ContDiff.differentiable`;
- mean value: `Convex.norm_image_sub_le_of_norm_fderiv_le`;
- measure: `Measure.addHaar_ball_of_pos` and `Metric.measure_ball_pos`;
- finiteness: `measure_ball_lt_top`;
- dimension: `Module.finrank_fintype_fun_eq_card`;

- set integrals: `setIntegral_ge_of_const_le` and `setIntegral_mono_set`;
- integrability: `ContinuousOn.integrableOn_compact`;
- thickenings: `IsCompact.cthickening` and `Metric.mem.cthickening_of_dist_le`;
- bounds: `IsCompact.exists_bound_of_continuousOn`;
- roots: `Real.rpow_inv_natCast_pow` and `Real.pow_rpow_inv_natCast`;
- monotonicity: `le_of_pow_le_pow_left0`.

Seabed: `superPoly_setIntegral_abs_exp_sub` and `superPoly_of_forall_eventually_le` from the total-variation tide.

New: the gradient lemmas, the generic peak lemma, the gradient bound on a compact, the powered and root inequalities, the uniform theorem and the pointwise corollary. About 340 lines.

6 Lessons

- For CLAUDE.md: `omit [Fintype iota] in` is refused as soon as a norm or derivative on $\iota \rightarrow \mathbb{R}$ appears in the statement, since the `Pi` normed instances need it; and `HasFDerivAt.sub` yields a `Pi`-form function, so name the derivative fact with the `lambda` type before rewriting with `.fderiv`.
- For the tide skill: an L^1 -to- L^∞ upgrade by a Lipschitz peak is a reusable pattern whenever the objects have polynomially bounded gradients; the generic peak lemma is the piece to reach for.
- For the paper: the closure theorems now hold at the level of the weights themselves, uniformly on compacts, not only against test functions. The $o(t^{-\infty})$ agreement is pointwise and locally uniform.

7 Follow-ups

- *Derivatives.* Uniform beyond-all-orders agreement of Dh_t (and all fixed-order derivatives) on compacts, by a second interpolation with $\|D^2h_t\| \leq B_K t^2$ and Taylor along segments with step $s = t^{-(N+3)}$; the consult's sketch is in the tide log.
- *Normalized densities.* $|e^{-tL_2}/Z_2 - e^{-tL_1}/Z_1| \leq Z_2^{-1}(|h_t| + |Z_2/Z_1 - 1|)$ gives uniform agreement of the normalized densities on compacts once the anchor bounds Z_i below.
- *Weaker regularity.* Restate (W) and (U) with C^1 losses of locally bounded gradient and a smooth difference, the only facts used.

Acknowledgements

GPT-6 Astra proposed the mechanism in the previous tide's consult, confirmed the constants, corrected the note on derivatives, and recommended the generic peak lemma. The seabed is

the day's germbij tides.